

What is a Data Analysis?

Turning raw facts (numbers, text, dates) **into patterns and answers that help you decide something.**

Where does the data live?

Data usually sits in a table — rows and columns, like a spreadsheet.

A **row** = one record, one event, one entry (e.g. one sale, one customer, one transaction).

A **column** = one type of information about that record (e.g. Country, Product, Revenue).

The two kinds of columns you'll always see

Dimension — a word or label. Describes or groups something (Country, Product, Date, Category). You use it to answer "by what?"

Metric — a number. Something you can count, sum, or average (Revenue, Units Sold, Profit). You use it to answer "how much / how many?"

The workflow

Collect → Clean → Analyze → Visualize → Decide

Where data analysis is used (industries)

Data analysis isn't tied to one field — almost every industry runs on it. A few examples:

- **Healthcare** — tracking patient outcomes, hospital wait times, disease spread, drug effectiveness
- **Retail & e-commerce** — sales trends, best-selling products, customer buying patterns (like AfriMart)
- **Finance & banking** — fraud detection, credit risk, investment performance, spending patterns
- **Marketing** — campaign performance, customer segmentation, ad spend vs. return
- **Sports** — player performance stats, team strategy, injury prediction
- **Government & public policy** — census data, crime rates, economic indicators
- **Manufacturing** — production efficiency, defect rates, supply chain tracking
- **Transportation & logistics** — delivery times, route optimization, fuel usage
- **Education** — student performance, dropout risk, course effectiveness
- **Tech & social media** — user engagement, app usage, recommendation systems

One sentence to remember this

Any place that generates numbers and needs better decisions — that's a place for data analysis.

What is a Dashboard?

A dashboard is a single page that shows important business information at a glance.

- ☐ Used by managers and decision-makers
- ☐ Focuses on insights, not raw data

Example:

Revenue, Profit, and Sales trends on one screen



AfriMart KollyBright Dashboard



Côte d'Ivoire

Egypt

Ghana

Kenya

Nigeria

Rwanda

Senegal

South Africa

Tanzania

Uganda

2024

2025

2026

<1/2/2024

>12/30/2026



Revenue

₦ 10,447,536,600



Profit

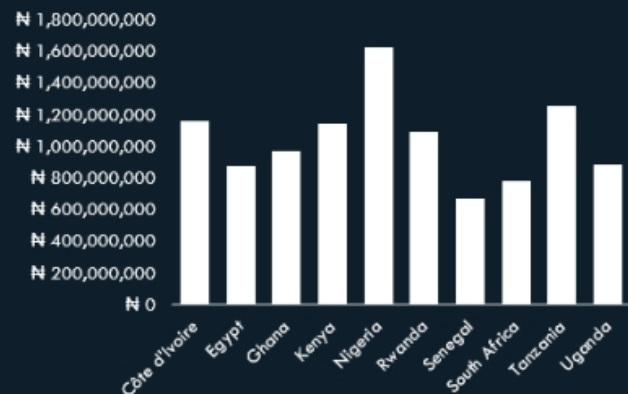
₦ 2,099,369,250



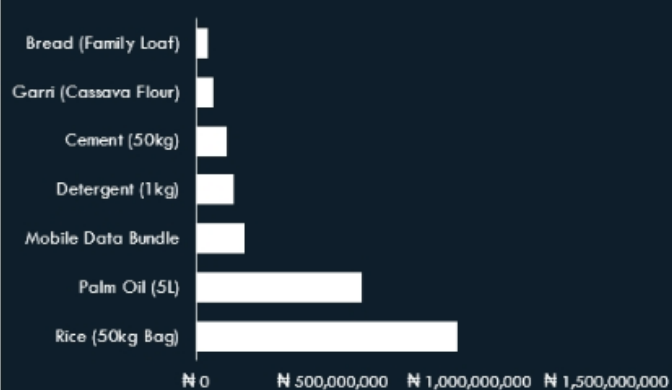
Unit Sold

1017885

Revenue by Country



Profit by Product



₦ 4,500,000,000

₦ 4,000,000,000

₦ 3,500,000,000

₦ 3,000,000,000

₦ 2,500,000,000

₦ 2,000,000,000

₦ 1,500,000,000

₦ 1,000,000,000

₦ 500,000,000

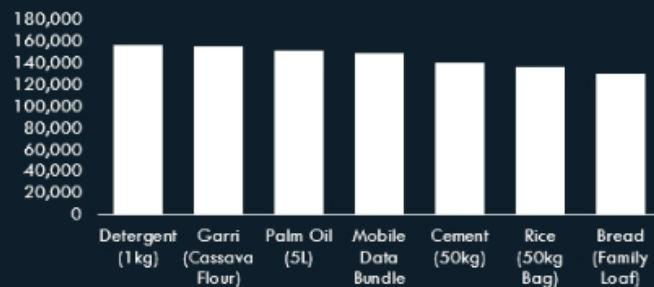
₦ 0

2024

2025

2026

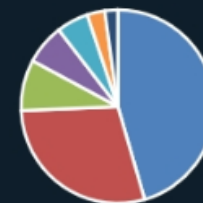
Unit Sold by Product



Sum of Revenue Sum of Units Sold



Top 5 Products



Rice (50kg Bag)

Palm Oil (5L)

Mobile Data Bundle

Detergent (1kg)

Cement (50kg)

Garri (Cassava Flour)

Bread (Family Loaf)



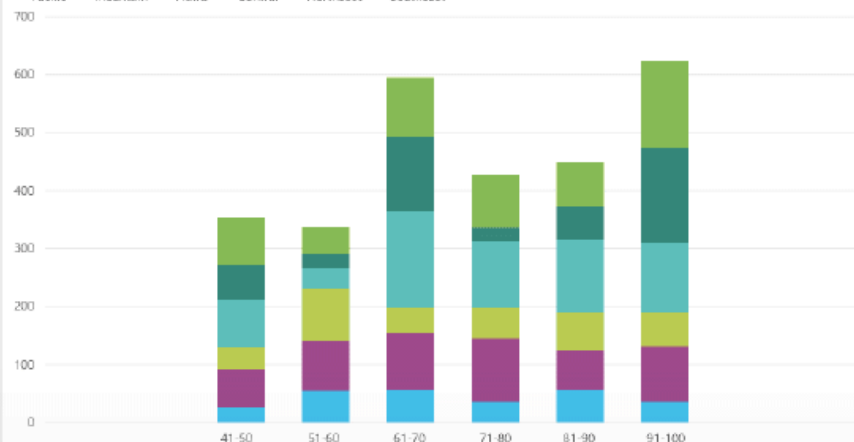
Travel Analysis

Ask a question about your data

Regional Load

BY TERRITORY

Pacific Mountain Plains Central Northeast Southeast



Delayed Flights

BY DATE



Actual Flight Expenditures, Bu...

BY TRAVEL MONTH



Budget Remaining

BY TRAVEL MONTH



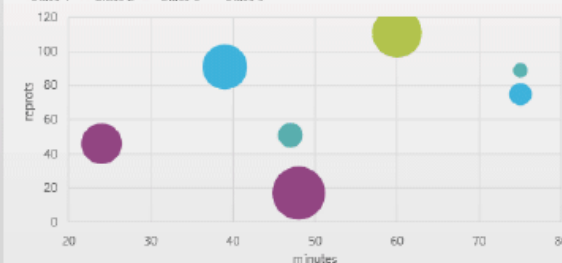
Flight Expense



Rush Booking, Trip Class, Season

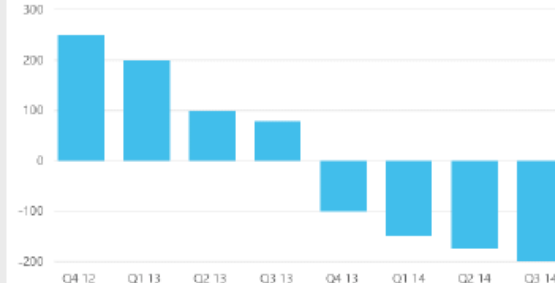
BY CATEGORY, TYPE

Class 1 Class 2 Class 3 Class 3



Sales

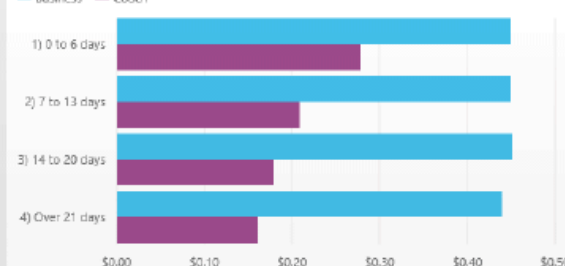
BY PERIOD



Average Cost Per Mile

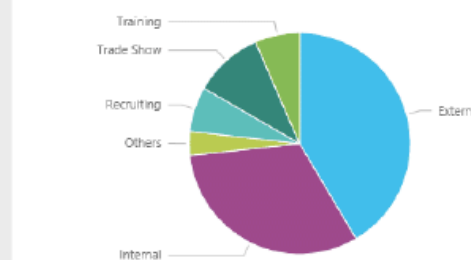
BY ADVANCED BOOKING CATEGORY, TRIP CLASS

Business Coach



Number of Trips

BY TRIP PURPOSE



On Time Arrival

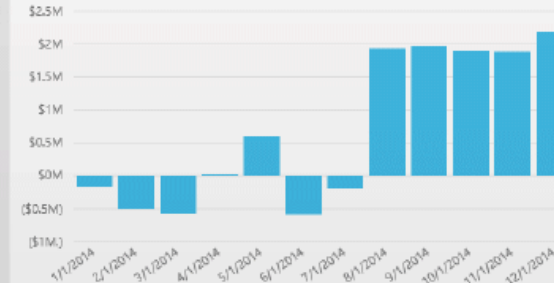
BY COUNTRY, YEAR

2013 2014



Budget Remaining

BY TRAVEL MONTH

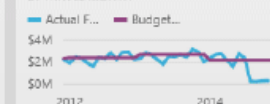


Variance to Budget

(\$1.39M)

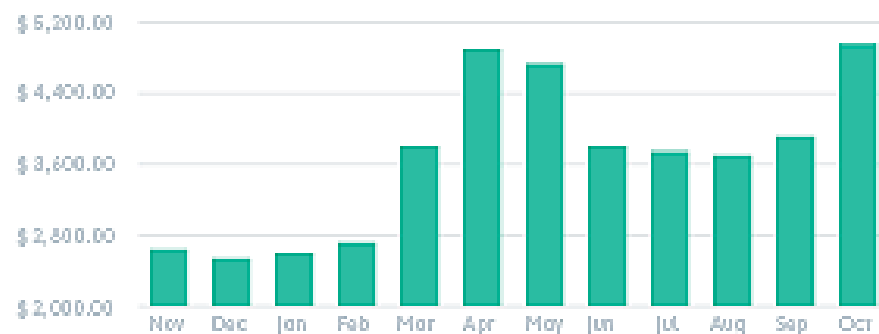
Actual Flight Expenditures, Bu...

BY TRAVEL MONTH

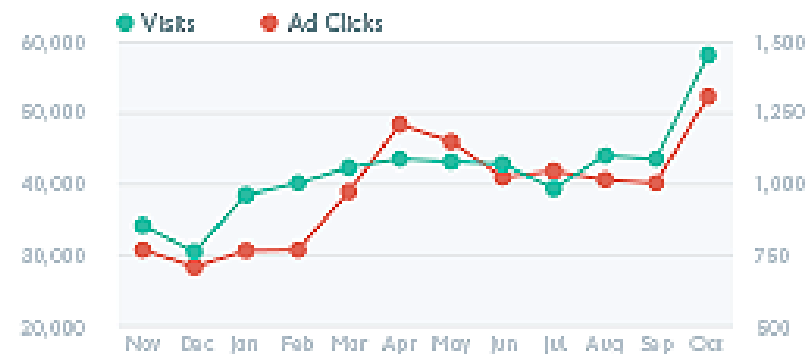


WIDGETS

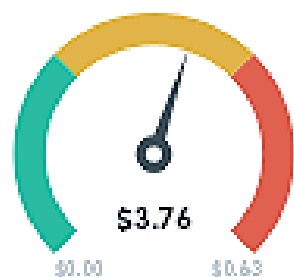
AD COST - TREND WIDGET



VISITS - TREND WIDGET 2 METRICS

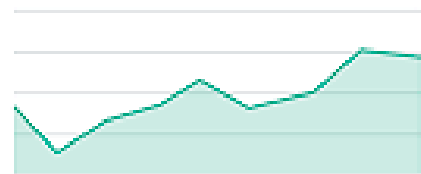


COST PER CLICK (CPC) - GAUGE WIDGET



AD IMPRESSIONS - HISTORIC WIDGET

35,567



Previous period

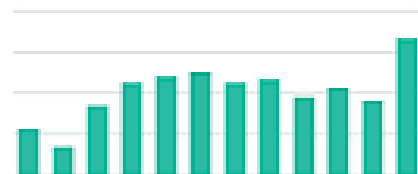
0 % ^

Previous year

3 % ^

TOTAL PAGEVIEWS

252,292



Previous period

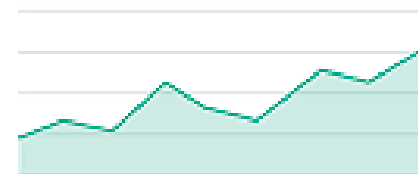
34 % ^

Previous year

62 % ^

TOTAL VISITS

57,841



Previous period

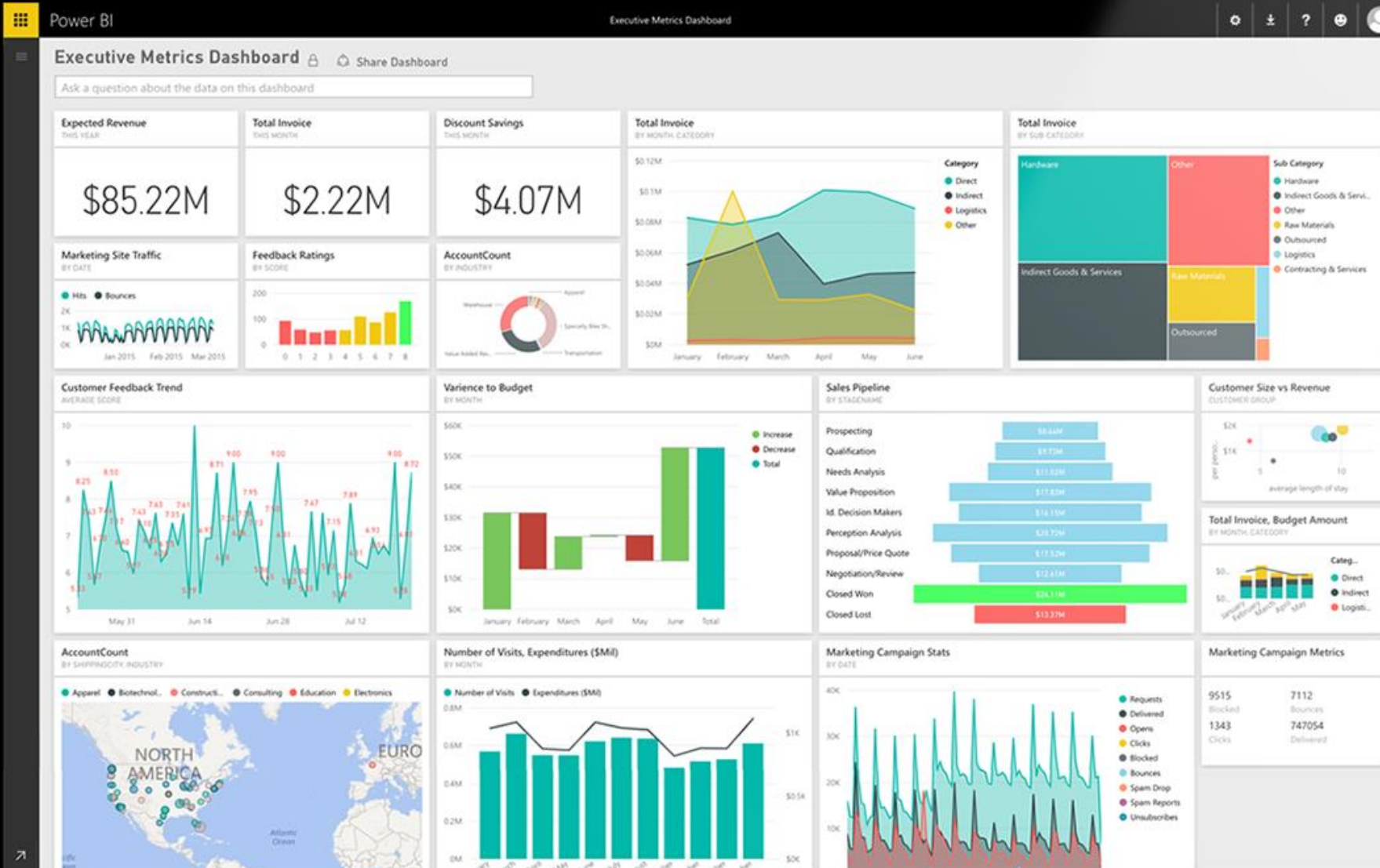
34 % ^

Previous year

62 % ^



Project Builders
Build Real Experience



12 dashboard design tips

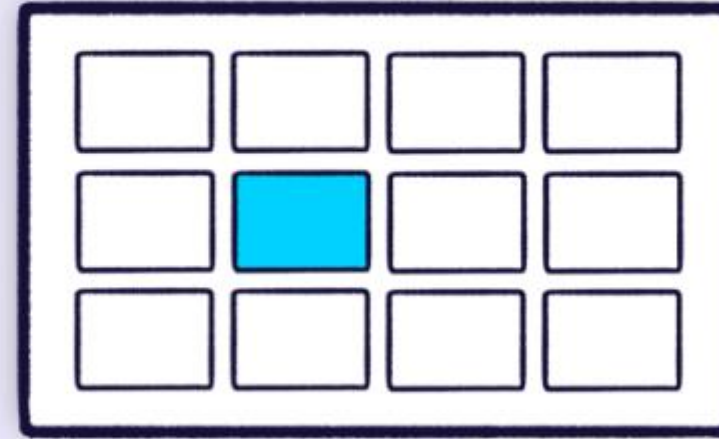
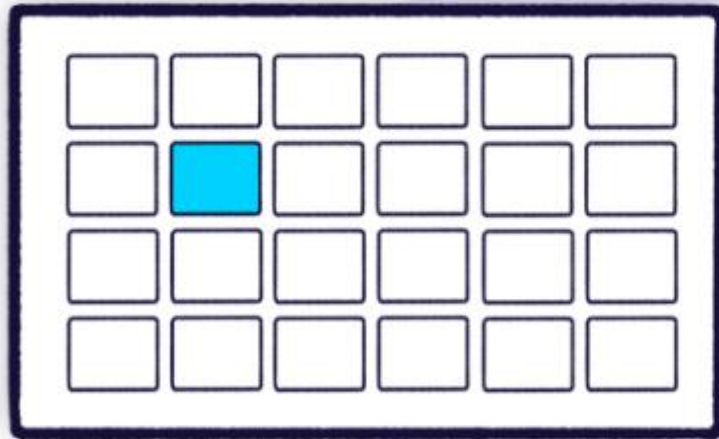
- 1 Be clear about what you're trying to achieve – your board's purpose will inform its design
- 2 Only include what's important – everything should support your board's intent
- 3 Consider data ink ratio – avoid decorative elements that don't communicate data
- 4 Round your numbers – being overly precise can get in the way of important changes
- 5 Use the most efficient visualization – a good visualization should be understood quickly
- 6 Group your related metrics – make your metrics easy to find
- 7 Be consistent – using the same visualizations and layouts makes comparing easier
- 8 Use size and position to show hierarchy – make it clear to the viewer what's most important
- 9 Give your numbers context – help your viewers know if a number is good, bad, normal or unusual
- 10 Use clear labels for your audience – keep them short and self explanatory
- 11 Remember it's for people – it's ok to break the rules if it increases engagement
- 12 Keep evolving your dashboards – check that your dashboard is encouraging the right behaviour

Be clear about what you're trying to achieve



PURPOSE

Include only the most important content



Round your numbers

\$19901

\$431.72

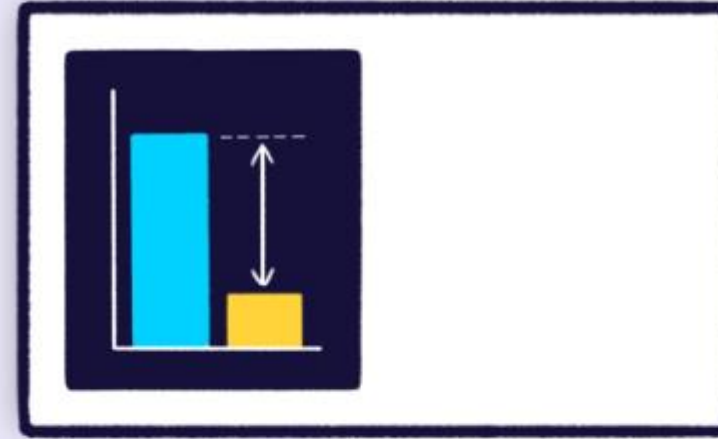
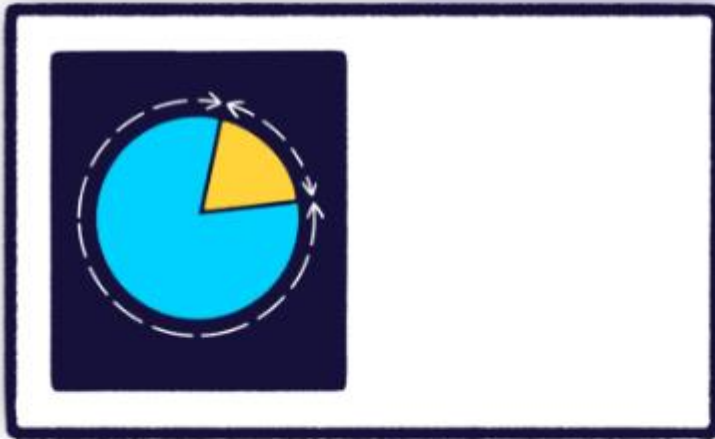


\$19.9k

\$431



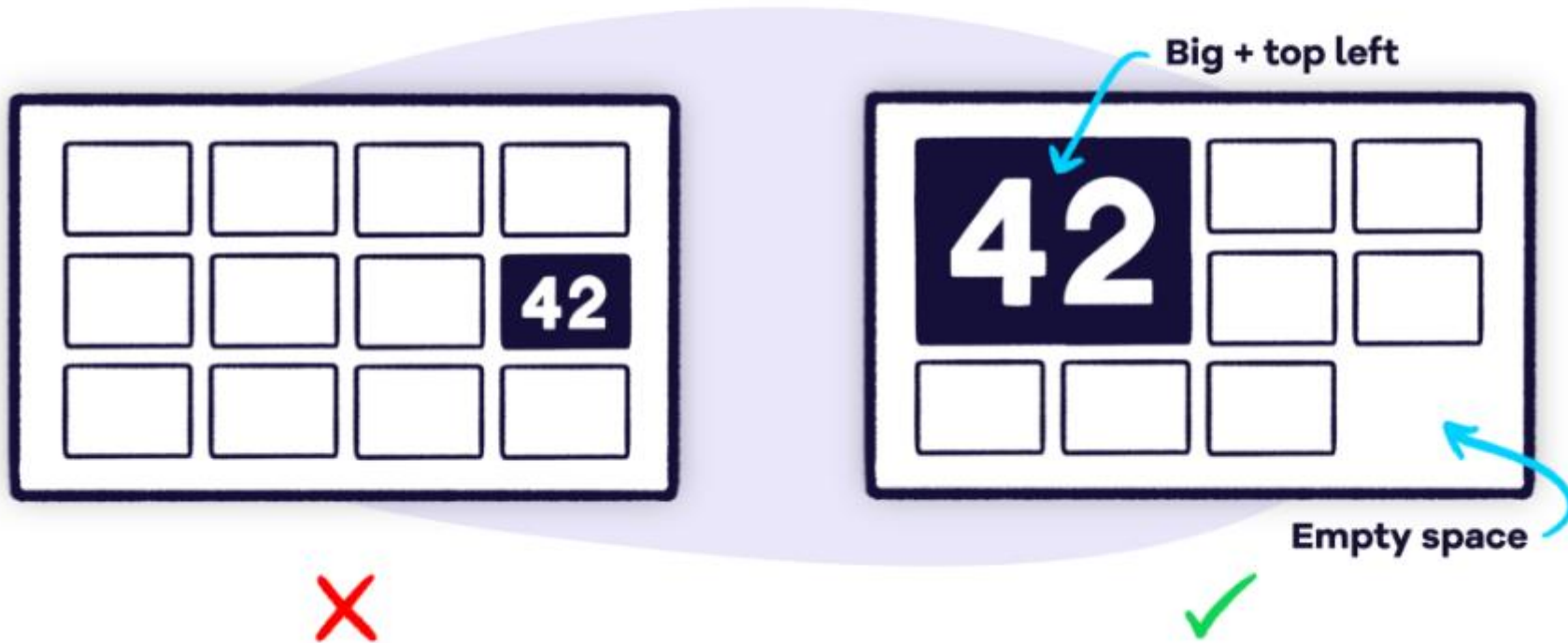
Use the most efficient visualization



Be consistent



Use size and position to show hierarchy



Give your numbers context

42



42

↑ 37%



Use clear labels your audience will understand

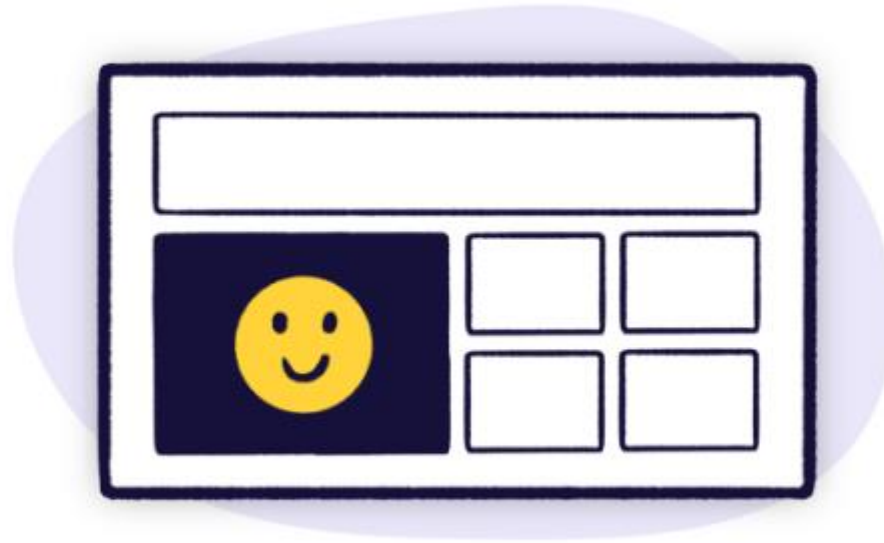
\$190k ?



Total revenue this year
\$190k



Remember it's for people



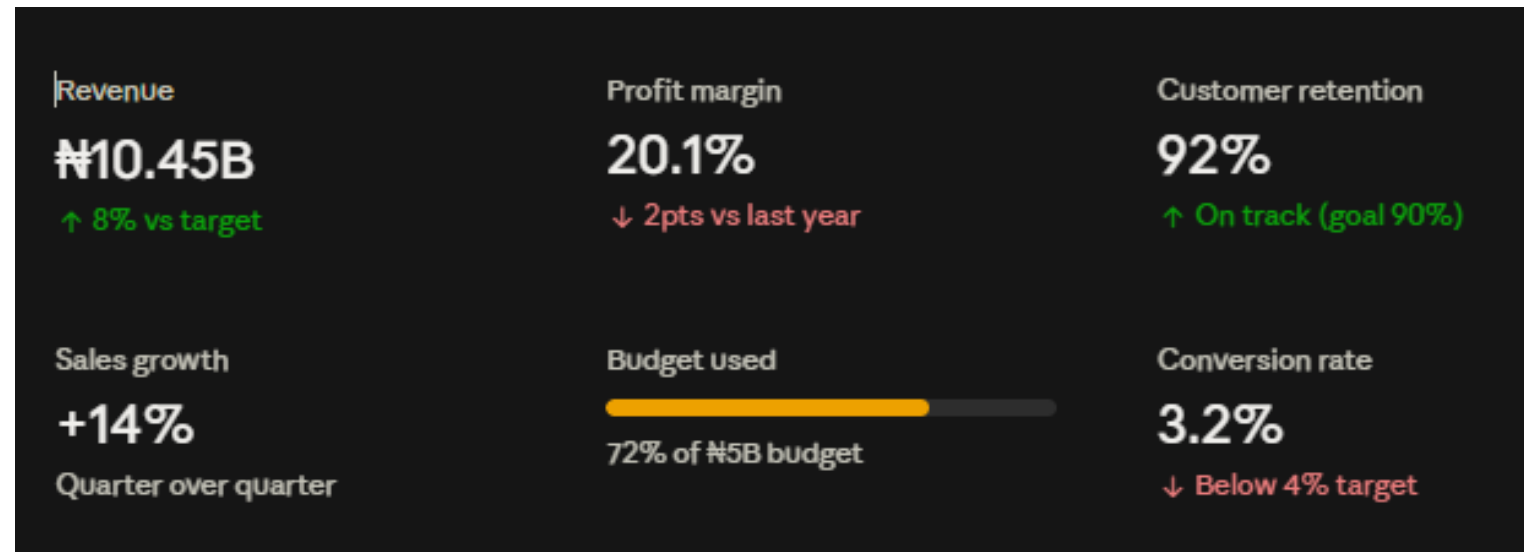
What is a KPI?

KPI (Key Performance Indicator) is a number that tells us **how well the business is doing**.

“KPIs answer the question: *Are we doing well or not?*”

Examples

- ✓ Total Revenue
- ✓ Total Profit
- ✓ Total Units Sold
- ✓ Profit Margin (%)



IMPORTANT NOTE

“KPIs are usually **big numbers at the top** of a dashboard.”



AfriMart KollyBright Dashboard



▼ Revenue
₦ 10,447,536,600

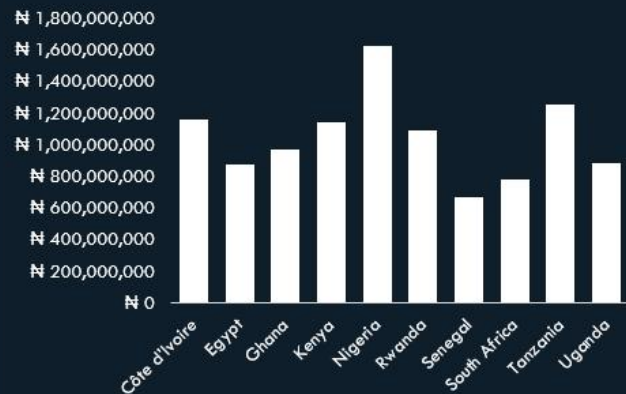


▲ Profit
₦ 2,099,369,250

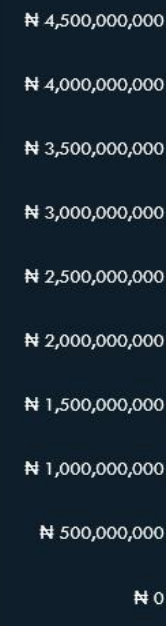
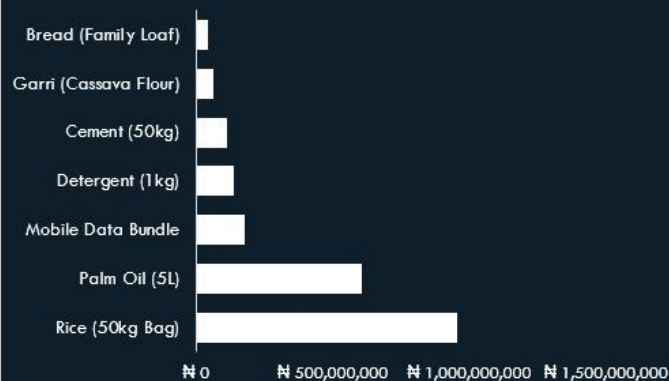


▲ Unit Sold
1017885

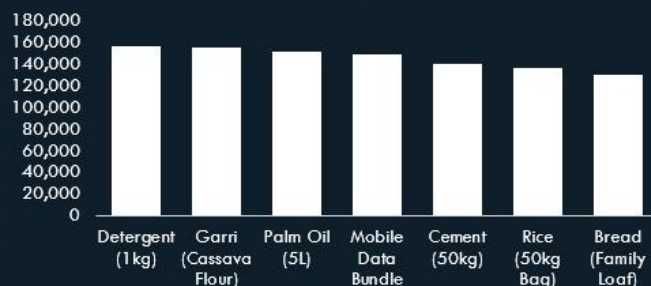
Revenue by Country



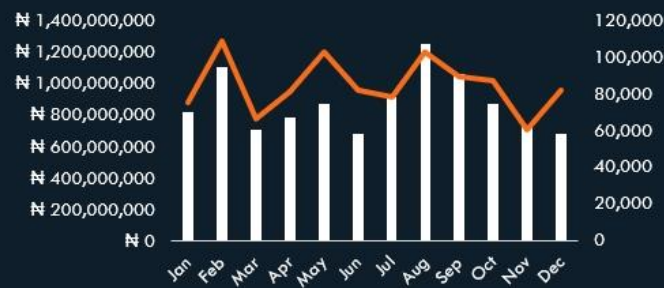
Profit by Product



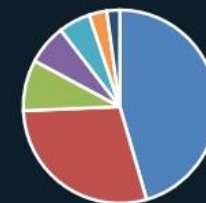
Unit Sold by Product



Sum of Revenue and Sum of Units Sold



Top 5 Products



■ Rice (50kg Bag) ■ Palm Oil (5L) ■ Mobile Data Bundle ■ Detergent (1kg)
■ Cement (50kg) ■ Garri (Cassava Flour) ■ Bread (Family Loaf)

Côte d'Ivoire

Egypt

Ghana

Kenya

Nigeria

Rwanda

Senegal

South Africa

Tanzania

Uganda

2024

2025

2026

<1/2/2024

>12/30/2026

Give your numbers context

42



42

↑ 37%



Metrics vs Dimensions

Metric

A **number** we calculate.

Examples:

- Revenue
- Profit
- Units Sold

Dimension

How we **group or break** down a metric.

Examples:

- Country
- Product
- Date

 Dimension Describes or labels data	 Metric Measures or counts data
Country: Nigeria	Revenue: ₦4.2B
Product: Shoes	Units sold: 210,000
Month: January	Profit margin: 20%

Metrics + Dimensions

Example:

Metric: Revenue

Dimension: Country

→ **Revenue by Country**

#####

Metric: Profit

Dimension: Product

→ **Profit by Product**

How to Name Charts Properly

Rule:

Metric by **Dimension**

Examples:

- ✓ **Sales** by **Country**
- ✓ **Revenue** by **Product**
- ✓ **Profit** by **Year**
- ✓ **Units Sold** by **Month**

When There Are Two Dimensions

If there are two categories, mention both.

Rule:

Metric by ***Dimension and Dimension***

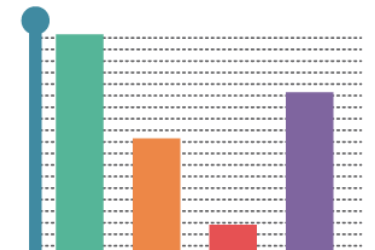
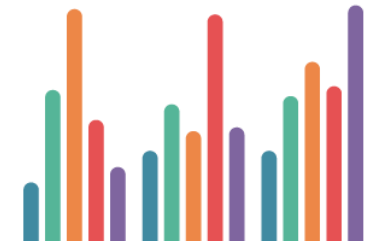
Examples:

- ✓ ***Revenue*** by ***Country and Product***
- ✓ ***Profit*** by ***Product and Year***
- ✓ ***Sales*** by ***Country and Year***

Why We Use Charts

Charts help us understand data quickly.

- ❖ Compare values
- ❖ Spot trends
- ❖ Identify top and bottom performers



Examples:

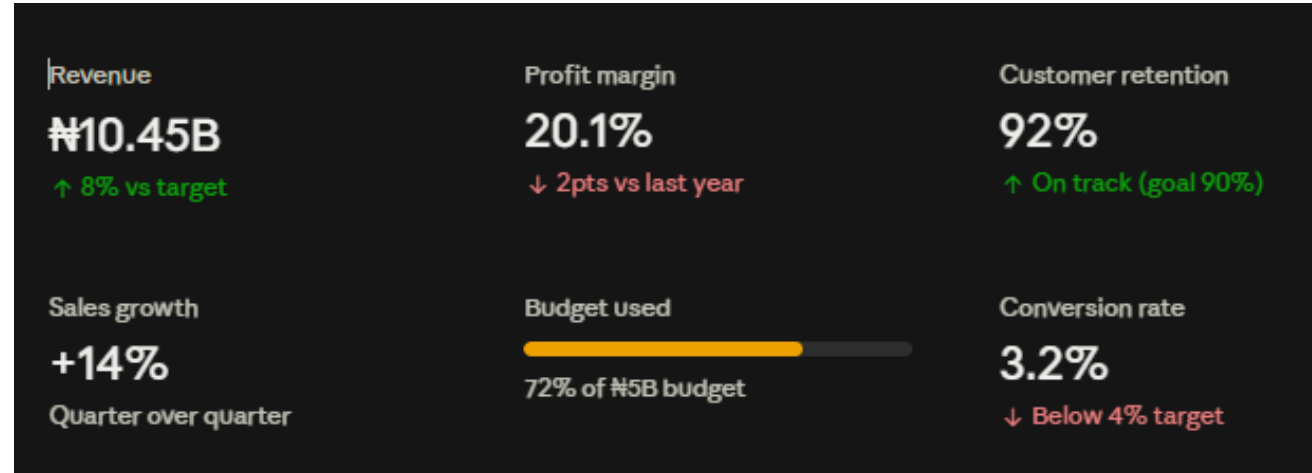
- **Bar chart** → Compare products
- **Column chart** → Compare countries
- **Line chart** → Show trends over time

Choose charts based on data type - use bar charts for comparisons, line charts for trends, and tables for detailed data. Avoid pie charts unless showing parts of a whole.

KPIs vs Charts

KPIs

- Show totals
- Big numbers
- Quick summary

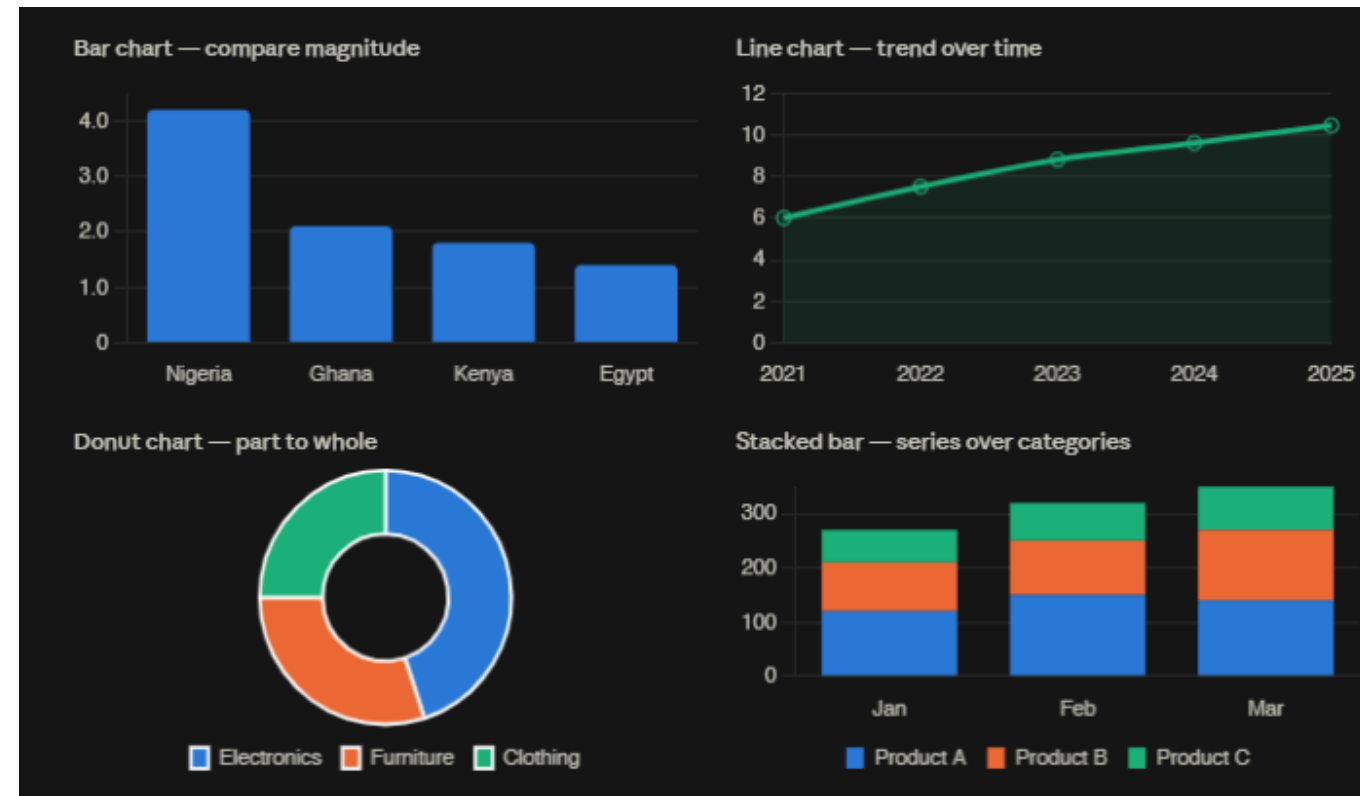


Charts

- Show breakdowns
- Explain the numbers
- Show comparisons

Example:

- KPI: Total Revenue
- Chart: Revenue by Country



One Chart, One Question

Each chart should answer one business question.

Example:

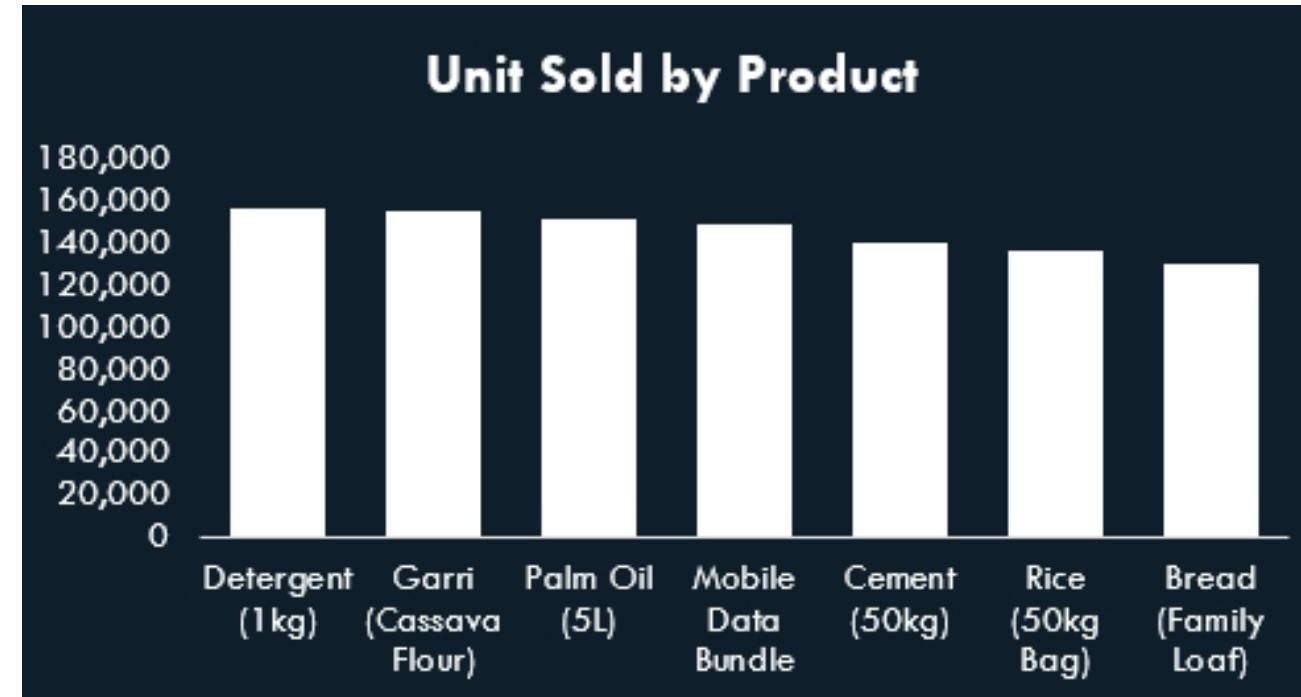
- **Question:** Which country makes the most money?
- **Chart:** Revenue by Country

COMMON DASHBOARD MISTAKES

Avoid:

- Too many charts
- Too many colors
- No clear titles
- No KPIs

Keep dashboards clean and simple.



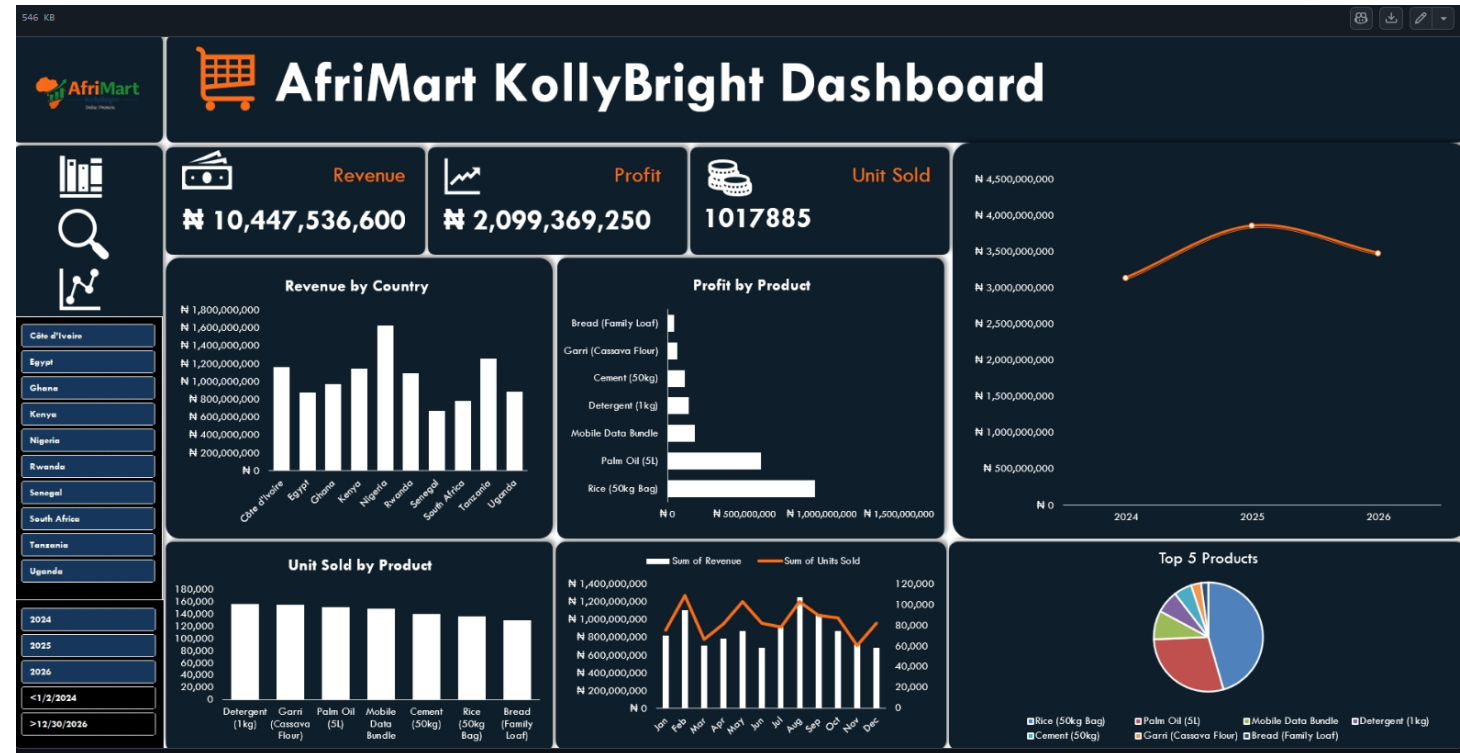
Dashboard Layout (Best Practice)

1. KPIs at the top
2. Charts in the middle
3. Filters (slicers) on the side
4. Clear titles everywhere

Key Takeaway

A good dashboard:

- Is simple
- Answers business questions
- Uses KPIs, charts, and slicers correctly
- Can be understood in seconds



[**Metric**] by [**Dimension(s)**]

Metric → the number being calculated (Revenue, Profit, Units Sold)

Dimension → how the number is broken down (Country, Product, Date)

<u>DATA USED</u>	<u>CORRECT NAME</u>
Sales + Country	Sales by Country
Revenue + Product	Revenue by Product
Profit + Product	Profit by Product
Units Sold + Country	Units Sold by Country

“We Always Start With The Number, Not The Category.”

AfriMart sales dataset — before we start



The scenario

AfriMart is a retail and distribution company operating across 10 African countries. **Each row in the dataset is one sales transaction** — a specific product, sold in a specific country, on a specific date, with the resulting money numbers. This is called transaction-level data: nothing is pre-summarized. Every chart or KPI you build is just these rows added up in different ways.

What's being sold

7 everyday retail goods — Rice, Palm Oil, Detergent, Bread, Garri, Cement, and Mobile Data Bundles — spanning January 2024 to December 2026 (3 years), across Nigeria, Egypt, Kenya, Ghana, South Africa, Uganda, Rwanda, Senegal, Tanzania, and Côte d'Ivoire.

Dataset structure (700 rows, 7 columns)

Column	Type	Role	What it means
Country	Text	Dimension	Which of the 10 countries the sale happened in
Product	Text	Dimension	Which of the 7 products was sold
Date	Date	Dimension	When the sale happened (Jan 2024 – Dec 2026)
Units Sold	Number	Metric	Quantity of product sold in that transaction
Revenue	Number	Metric	Money earned from the sale
Cost	Number	Metric	Cost of acquiring/producing the goods sold
Profit	Number	Metric	Revenue minus Cost

The one formula that explains every number

$Revenue - Cost = Profit$

- **Making the Dashboard dynamic**
- **Dashboard Structure**
- **Creating the charts and visuals**
- **Customising the visuals and putting it all together**

- ✓ **Left sidebar** → Navigation
- ✓ **Top bar** → Title / filters
- ✓ **Main content** → Charts, KPIs
- ✓ **Footer** → Notes / status

Dashboard Questions / Insights (Up to 8)

These are **realistic business questions** your dashboard can answer. Each can be visualized with a pivot table + chart:

Revenue by Country

Question: Which countries generate the most revenue?

Pivot: Rows = Country, Values = Revenue (Sum)

Chart: Column chart

Profit by Product

Question: Which products are most profitable?

Pivot: Rows = Product, Values = Profit (Sum)

Chart: Bar chart

Units Sold by Product

Question: Which products sell the most units?

Pivot: Rows = Product, Values = Units Sold (Sum)

Chart: Column chart

Revenue Trend Over Time

Question: How has revenue changed month to month or year to year?

Pivot: Rows = Date (group by Month/Year), Values = Revenue (Sum)

Chart: Line chart

Profit by Country

Question: Which countries are most profitable?

Pivot: Rows = Country, Values = Profit (Sum)

Chart: Column chart

Revenue by Country and Product

Question: Which products sell best in which countries?

Pivot: Rows = Country, Columns = Product, Values = Revenue (Sum)

Chart: Stacked column chart

Monthly Sales Performance

Question: Are there seasonal trends or peak sales months?

Pivot: Rows = Date (group by Month), Values = Revenue / Units Sold

Chart: Line or area chart

Top 5 Products by Profit

Question: Which 5 products generate the most profit?

Pivot: Rows = Product, Values = Profit, Sort descending, Top 5 filter

Chart: Bar chart

Top Section: KPIs

- Total Revenue
- Total Profit
- Total Units Sold
- Profit Margin (%)

Middle Section: Charts answering questions

- Revenue by Country
- Profit by Product
- Units Sold by Product
- Revenue Trend Over Time

Bottom / Side Section: Optional deeper analysis

- Revenue by Country and Product
- Top 5 Products by Profit
- Monthly Sales Performance

Slicers: Country, Product, Year



Before creating any dashboard, ask:

1. Who is the audience?

Executives, marketing, operations? Different users care about different metrics.

2. What questions need answering?

Only create visuals that answer business questions.

3. Which KPIs are critical?

Don't overload — 3–5 key metrics is enough.

4. How will interactivity help?

Use slicers for Country, Product, Year, etc.

KPIs (Up to 4)

KPIs are **big numbers at the top of your dashboard** that quickly tell the audience how the company is doing.

Based on your dataset (Country, Product, Units Sold, Revenue, Cost, Profit, Date), here are 4 KPIs you can use:

1. Total Revenue → Sum of Revenue across all countries and products

"Shows the total sales value for the business."

2. Total Profit → Sum of Profit across all countries and products

"Shows the money actually made after costs."

3. Total Units Sold → Sum of Units Sold

"Shows total volume of products sold."

4. Profit Margin (%) → Total Profit ÷ Total Revenue

"Shows how profitable the business is overall."

PART 2

Creating our Dashboard



AfriMart KollyBright Dashboard



▼ Revenue
₦ 10,447,536,600

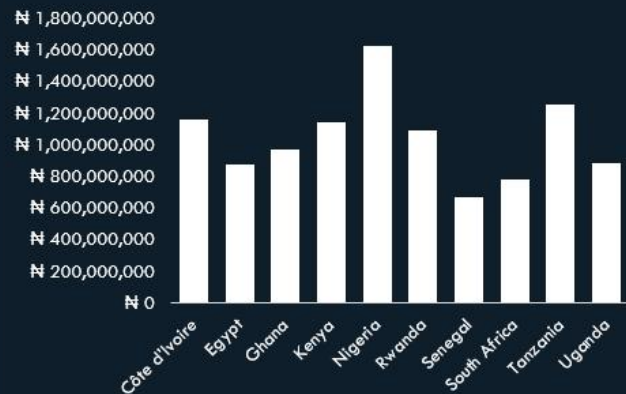


▲ Profit
₦ 2,099,369,250

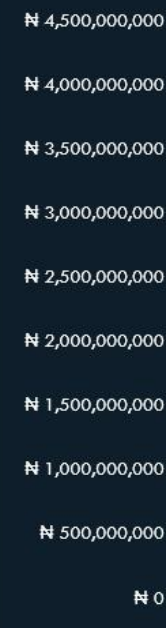
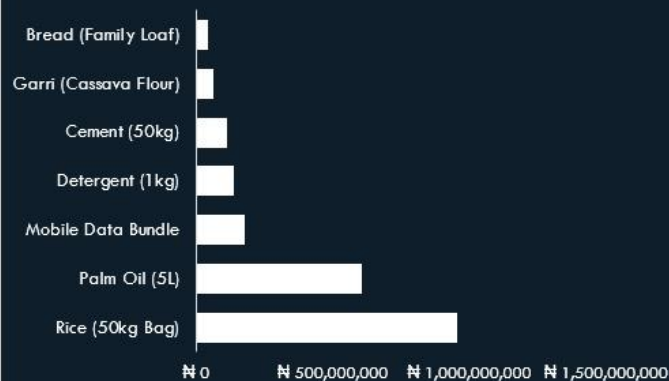


▲ Unit Sold
1017885

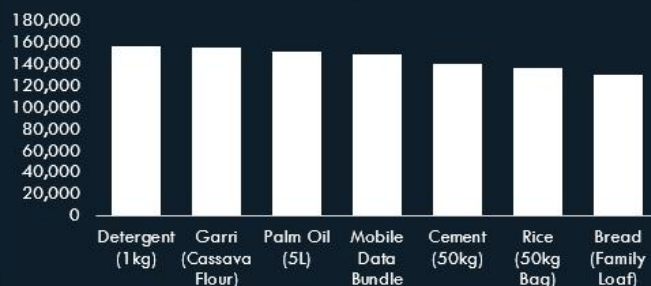
Revenue by Country



Profit by Product



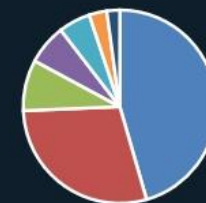
Unit Sold by Product



Sum of Revenue Sum of Units Sold



Top 5 Products



■ Rice (50kg Bag) ■ Palm Oil (5L) ■ Mobile Data Bundle ■ Detergent (1kg)
■ Cement (50kg) ■ Garri (Cassava Flour) ■ Bread (Family Loaf)

Côte d'Ivoire

Egypt

Ghana

Kenya

Nigeria

Rwanda

Senegal

South Africa

Tanzania

Uganda

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